

Lecture 5

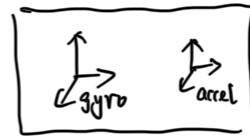
The IMU model and IMU integration:

This is one of the most important sensor for Robotics.

The readings from IMU:

$\{I\}$: IMU frame

$\{G\}$: global frame



Gyroscope: read angular velocity

Accelerometer: read acceleration

$$I_{wm} = I_w + b_g + n_g \quad (1)$$

I_w : true angular reading
 b_g : gyro bias, $b_g = \Pi_w g$, random walk
 $n_g \sim N(0, \sigma_g^2 I_3)$, Gaussian noises

$$I_{am} = I_a + b_a + n_a \quad (2)$$

I_a : true acceleration reading
 b_a : acce biases, $b_a = \Pi_w a$, random walk
 $n_a \sim N(0, \sigma_a^2 I_3)$, Gaussian noises

Another way of modeling accel:

$$I_{am} = (I_{ab} - \frac{\partial R}{\partial g} g) + b_a + n_a \quad g = \begin{bmatrix} 0 \\ 0 \\ -9.8 \end{bmatrix}$$

I_{ab} : body accel
 g : is gravity in G

We use (1) and (2) as our measurements.

$$\begin{cases} \text{True gyro reading: } I_w = I_{wm} - b_g - n_g \\ \text{True accel reading: } I_a = I_{am} - b_a - n_a \end{cases}$$

... IMU parameters:

when we model IMU, we use the following parameters:

$\{q_R, p_R\}$: IMU orientation and position: 6DOF

v_R : IMU velocity: 3DOF

b_g : gyro biases: 3DOF

b_a : accel biases: 3DOF

Then: our state vector for IMU: $x_I = \{q_R, p_R, v_R, b_g, b_a\}$: 15DOF

some place: use: quaternion: $q_R \leftrightarrow \frac{q}{2}$, then: $x_I = \begin{bmatrix} \frac{q}{2}^T & q_R^T & v_R^T & b_g^T & b_a^T \end{bmatrix}^T$
 $16 \times 1 \Rightarrow 15 \text{ dof}$

The continuous and discrete noise.

For IMU: we give continuous noise: $n_c \sim \mathcal{N}(0, \sigma_c^2 I_3)$

$$E(n_c(t)) = 0_{3 \times 1}$$

$$E(n_c(t+\tau)n_c^T(t)) = \sigma_c^2 I_3 \delta(\tau) \quad \delta(\tau) \text{ is impulse function: } \begin{cases} \tau=0 & \delta(\tau)=1 \\ \tau \neq 0 & \delta(\tau)=0 \end{cases}$$

For: $b'(t) = n_c(t)$, $n_c \sim \mathcal{N}(0, \sigma_c^2 I_3)$

if from t_k to t_{k+1} , then:

$$\int_{t_k}^{t_{k+1}} b'(t) dt = \int_{t_k}^{t_{k+1}} n_c(t) dt$$

$$= b_{k+1} - b_k$$

$$\Delta b = b_{k+1} - b_k = \int_{t_k}^{t_{k+1}} n_c(t) dt$$

$$E(\Delta b) = E\left(\int_{t_k}^{t_{k+1}} n_c(t) dt\right) = 0_{3 \times 1}$$

$$\begin{aligned} \text{cov}(\Delta b) &\triangleq P_{\Delta b} = E(\Delta b \Delta b^T) = E\left(\int_{t_k}^{t_{k+1}} n_c(t) dt \cdot \int_{t_k}^{t_{k+1}} n_c^T(t) dt\right) \\ &= E\left(\int_{t_k}^{t_{k+1}} \int_{t_k}^{t_{k+1}} n_c(t) n_c^T(\tau) dt d\tau\right) \\ &= E\int_{t_k}^{t_{k+1}} n_c(t) n_c^T(t) dt \end{aligned}$$

$$= \int_{t_k}^{t_{k+1}} E(n_c(t) n_c^T(t)) dt$$

$$= \int_{t_k}^{t_{k+1}} \sigma_c^2 I_3 dt = \delta t \sigma_c^2 I_3$$

Hence, we can discretize the n_c in $[t_k, t_{k+1}]$ as: $n_d \sim N(0, \frac{\sigma_c^2}{\delta t})$ $\delta t = t_{k+1} - t_k$

Then: $\Delta b = b_{k+1} - b_k = \int_{t_k}^{t_{k+1}} n_c dt = \int_{t_k}^{t_{k+1}} dt \cdot n_d = n_d \cdot \delta t$ $\delta t = t_{k+1} - t_k$

Hence: $\text{cov}(\Delta b) = E(\Delta b \Delta b^T) = \delta t^2 \cdot E(n_d n_d^T) = \delta t^2 \cdot \frac{\sigma_c^2}{\delta t} I_3 = \frac{\sigma_c^2}{\delta t} I_3 = \text{cov}(\Delta b)_c$

The first order ODE: ordinary differential equation

$$\begin{cases} \dot{x}_2 = \frac{g}{2R} L^2 w - b_g - n_g \cdot x \\ \dot{p}_2 = g v_2 \\ \dot{v}_2 = g_a + g_g = \frac{g}{2R} z_a + g_g = \frac{g}{2R} (z_{am} - b_a - n_a) + g_g \\ \dot{b}_g = n_{wg}, \quad \dot{b}_a = n_{wa} \end{cases}$$

Then:

$$\begin{cases} \dot{x}_2 = \frac{g}{2R} L^2 w - b_g - n_g \cdot x \\ \dot{p}_2 = g v_2 \\ \dot{v}_2 = \frac{g}{2R} (z_{am} - b_a - n_a) + g_g \\ \dot{b}_g = n_{wg} \\ \dot{b}_a = n_{wa} \end{cases}$$

$$\dot{x}_2 = f(x_2, n_2)$$

with $n_2 = \begin{bmatrix} n_g \\ n_a \\ n_{wg} \\ n_{wa} \end{bmatrix}$

discretized noise $n_{2d} = \begin{bmatrix} n_{gd} \\ n_{ad} \\ n_{wgd} \\ n_{wad} \end{bmatrix}$

$n_g \sim N(0, \sigma_g^2 I_3)$
 $n_a \sim N(0, \sigma_a^2 I_3)$
 $n_{wg} \sim N(0, \sigma_{wg}^2 I_3)$
 $n_{wa} \sim N(0, \sigma_{wa}^2 I_3)$

$\sigma_{gd}^2 = \sigma_g^2 / \delta t$
 $\sigma_{ad}^2 = \sigma_a^2 / \delta t$
 $\sigma_{wgd}^2 = \sigma_{wg}^2 / \delta t$
 $\sigma_{wad}^2 = \sigma_{wa}^2 / \delta t$

The IMU model is done:

There are various IMU models, but the basic idea is the same.

The IMU Integration:

Big picture: given a typical linear system: from lecture 1:

(VINS)
Visual-Inertial Navigation System

state evolution $\rightarrow$ IMU model: $\rightarrow \dot{x}_2 = f(x_2, n_2)$ ②

$$\begin{cases} x = Ax + Bu & \rightarrow \text{state transition} \\ y = Cx + Du & \rightarrow \text{observation / measurement} \end{cases} \rightarrow \text{vision / camera model} \rightarrow \boxed{z_c = h_c(x_2) + n_c} \quad (4)$$

③ and ④ will our focus for our tutorials.

How to process ④ is covered in lecture 3 and lecture 4.

We will learn how to process IMU in this lecture 5.

The problem formulation:

Given $x_k \sim N(\hat{x}_k, P_k)$, IMU reading (z_{wmk}, z_{amk}) ,

solve: $x_{k+1} \sim N(\hat{x}_{k+1}, P_{k+1}) \Rightarrow$ Two targets: $\begin{cases} \hat{x}_{k+1} \\ P_{k+1} \end{cases}$

$$\underbrace{x_k \sim N(\hat{x}_k, P_k)}_{\text{current state distribution}} \xrightarrow{\hat{x}_2 = f(x_2, n_2)} \underbrace{x_{k+1} \sim N(\hat{x}_{k+1}, P_{k+1})}_{\text{next state distribution}}$$

How to solve it?:

From $\hat{x}_2 = f(x_2, n_2)$, we can get: $x_{k+1} = \overset{\text{integrated function}}{f_{\text{int}}}(x_k, z_{wm}, z_{am}, n_k)$. $\begin{cases} x_{k+1} = x_{I_{k+1}} \\ x_k = x_{I_k} \\ n_k = n_{I_k} \sim N(0, R_{I_k}) \end{cases}$

Hence: $\hat{x}_{k+1} = f_{\text{int}}(\hat{x}_k, z_{wm}, z_{am}, 0) \rightarrow \text{solved } \hat{x}_{k+1}$

Then, linearize it, we can get: $\tilde{x}_{k+1} = \Phi_{(k+1,k)} \tilde{x}_k + G_k n_{kd}^*$ n_{kd} : is discretized noise

Then: $P_{k+1} = E(\tilde{x}_{k+1} \tilde{x}_{k+1}^T)$

$$P_{k+1} = \Phi_{(k+1,k)} P_k \Phi_{(k+1,k)}^T + G_k R_{I_d} G_k^T \rightarrow \text{solved } P_{k+1}$$

Given the big picture, let's explore the detailed derivations.

derive:

$$\begin{aligned} \text{The input: } \mathbf{z}_w &= \mathbf{z}_{wm} - \mathbf{b}_g - \mathbf{n}_g = \underbrace{\mathbf{z}_{wm} - \hat{\mathbf{b}}_g}_{\mathbf{z}_{\hat{w}}} - \underbrace{\tilde{\mathbf{b}}_g - \mathbf{n}_g}_{\mathbf{z}_{\tilde{w}}} \\ &= \mathbf{z}_{\hat{w}} + \mathbf{z}_{\tilde{w}} \end{aligned}$$

$$\begin{aligned} \mathbf{z}_a &= \mathbf{z}_{am} - \mathbf{b}_a - \mathbf{n}_a = \underbrace{\mathbf{z}_{am} - \hat{\mathbf{b}}_a}_{\mathbf{z}_{\hat{a}}} - \underbrace{\tilde{\mathbf{b}}_a - \mathbf{n}_a}_{\mathbf{z}_{\tilde{a}}} \\ &= \mathbf{z}_{\hat{a}} + \mathbf{z}_{\tilde{a}} \end{aligned}$$

From k to $k+1$, $\Delta t = t_{k+1} - t_k$

For rotation:

$$\mathbf{G}_{\mathbf{z}_{k+1}} \mathbf{R} = \mathbf{G}_{\mathbf{z}_k} \mathbf{R} \cdot \underbrace{\frac{\mathbf{z}_k}{\mathbf{z}_{k+1}} \mathbf{R}}_{\Delta \mathbf{R}} \quad \Delta \mathbf{R} \stackrel{\Delta}{=} \frac{\mathbf{z}_k}{\mathbf{z}_{k+1}} \mathbf{R} \quad \Delta \mathbf{R} \text{ is rotation integration from } \mathbf{z}_w$$

For velocity

$$\mathbf{G}_{\dot{\mathbf{z}}_k} = \mathbf{G}_{\mathbf{z}_k} \mathbf{z}_k \mathbf{a} + \mathbf{G}_g \Rightarrow \int_{t_k}^{t_{k+1}} \mathbf{G}_{\dot{\mathbf{z}}_t} dt = \int_{t_k}^{t_{k+1}} (\mathbf{G}_{\mathbf{z}_t} \mathbf{z}_t \mathbf{a} + \mathbf{G}_g) dt$$

$$\text{LHS: } \int_{t_k}^{t_{k+1}} \mathbf{G}_{\dot{\mathbf{z}}_t} dt = \mathbf{G}_{\mathbf{z}_{k+1}} - \mathbf{G}_{\mathbf{z}_k}$$

$$\begin{aligned} \text{RHS: } \int_{t_k}^{t_{k+1}} (\mathbf{G}_{\mathbf{z}_t} \mathbf{z}_t \mathbf{a} + \mathbf{G}_g) dt &= \int_{t_k}^{t_{k+1}} \mathbf{G}_{\mathbf{z}_t} \mathbf{z}_t \mathbf{a} dt + \mathbf{G}_g \Delta t \\ &= \int_{t_k}^{t_{k+1}} \mathbf{G}_{\mathbf{z}_t} \mathbf{z}_t \frac{\mathbf{z}_k}{\mathbf{z}_t} \mathbf{R} \mathbf{z}_t \mathbf{a} dt + \mathbf{G}_g \Delta t \\ &= \mathbf{G}_{\mathbf{z}_k} \mathbf{R} \underbrace{\int_{t_k}^{t_{k+1}} \frac{\mathbf{z}_k}{\mathbf{z}_t} \mathbf{R} \mathbf{z}_t \mathbf{a} dt}_{\Delta \mathbf{V}} + \mathbf{G}_g \Delta t \end{aligned}$$

$$\begin{aligned} \text{Then: } \mathbf{G}_{\mathbf{z}_{k+1}} &= \mathbf{G}_{\mathbf{z}_k} + \mathbf{G}_{\mathbf{z}_k} \mathbf{R} \Delta \mathbf{V} + \mathbf{G}_g \Delta t \quad \Delta \mathbf{V} \stackrel{\Delta}{=} \int_{t_k}^{t_{k+1}} \frac{\mathbf{z}_k}{\mathbf{z}_t} \mathbf{R} \mathbf{z}_t \mathbf{a} dt \\ &\quad \Delta \mathbf{V} \text{ is integration of } \mathbf{z}_w \text{ and } \mathbf{z}_a \end{aligned}$$

For position:

$$\mathbf{G}_{\dot{\mathbf{p}}_{\mathbf{z}_k}} = \mathbf{G}_{\mathbf{v}_{\mathbf{z}_k}} \Rightarrow \int_{t_k}^{t_{k+1}} \mathbf{G}_{\dot{\mathbf{p}}_{\mathbf{z}_t}} dt = \int_{t_k}^{t_{k+1}} \mathbf{G}_{\mathbf{v}_{\mathbf{z}_t}} dt$$

$$\text{note that: } \mathbf{G}_{\mathbf{v}_{\mathbf{z}_t}} = \mathbf{G}_{\mathbf{v}_{\mathbf{z}_k}} + \mathbf{G}_{\mathbf{z}_k} \mathbf{R} \int_{t_k}^t \frac{\mathbf{z}_k}{\mathbf{z}_s} \mathbf{R} \mathbf{z}_s \mathbf{a} ds + \mathbf{G}_g \delta t \quad \delta t = t - t_k$$

$$\text{LHS} = \int_{t_k}^{t_{k+1}} \dot{q}_{p_{2k}} d\tau = q_{p_{2k+1}} - q_{p_{2k}}$$

$$\text{RHS} = \int_{t_k}^{t_{k+1}} q_{v_{2k}} d\tau:$$

$$= \int_{t_k}^{t_{k+1}} \left(q_{v_{2k}} + \frac{q}{z_k R} \int_{t_k}^{\tau} \frac{z_k}{z_s} R \dot{z}_s a ds + g \delta \tau \right) d\tau$$

$$= \int_{t_k}^{t_{k+1}} q_{v_{2k}} + g \delta \tau d\tau + \frac{q}{z_k R} \int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} \frac{z_k}{z_s} R \dot{z}_s a ds d\tau$$

$$= q_{v_{2k}} \delta t + \frac{1}{2} g \delta t^2 + \frac{q}{z_k R} \underbrace{\int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} \frac{z_k}{z_s} R \dot{z}_s a ds d\tau}_{\Delta P}$$

$$\text{Then: } q_{p_{2k+1}} = q_{p_{2k}} + q_{v_{2k}} \delta t + \frac{q}{z_k R} \Delta P + \frac{1}{2} g \delta t^2 \quad \Delta P = \int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} \frac{z_k}{z_s} R \dot{z}_s a ds d\tau$$

is double integration of $\dot{z}_w$ and $\dot{z}_a$

$$\text{For gyro bias: } b_g = nwg \Rightarrow \int_{t_k}^{t_{k+1}} b_g d\tau = \int_{t_k}^{t_{k+1}} nwg d\tau$$

$$b_{gk+1} = b_{gk} + \int_{t_k}^{t_{k+1}} nwg d\tau$$

$$\text{For accel bias: } b_{ak+1} = b_{ak} + \int_{t_k}^{t_{k+1}} nwad\tau$$

Finally:

$$\left\{ \begin{array}{l} \frac{q}{z_{k+1}} R = \frac{q}{z_k} R \cdot \Delta R \\ q_{p_{2k+1}} = q_{p_{2k}} + q_{v_{2k}} \delta t + \frac{q}{z_k R} \Delta P + \frac{1}{2} g \delta t^2 \\ q_{v_{2k+1}} = q_{v_{2k}} + \frac{q}{z_k R} \Delta V + g \delta t \\ b_{gk+1} = b_{gk} + \int_{t_k}^{t_{k+1}} nwg d\tau \\ b_{ak+1} = b_{ak} + \int_{t_k}^{t_{k+1}} nwad\tau \end{array} \right.$$

$$\Delta R = \frac{z_k}{z_{k+1}} R$$

$$\Delta P = \int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} \frac{z_k}{z_s} R \dot{z}_s a ds d\tau$$

$$\Delta V = \int_{t_k}^{t_{k+1}} \frac{z_k}{z_s} R \dot{z}_s a d\tau$$

$$\Delta b_g = \int_{t_k}^{t_{k+1}} nwg d\tau$$

$$\Delta b_a = \int_{t_k}^{t_{k+1}} nwad\tau$$

$$x_{k+1} = f_{\text{int}}(x_k, z_{w_m}, z_{a_m}, n_k)$$

To this step: no assumption or approximation is made. Every step is accurate.
 z_{w_m} and z_{a_m} decides by the integration

The way how to compute $\Delta R, \Delta P, \Delta V$ given $\dots$

v

algorithms:

A few discussions:

① $\Delta R, \Delta P, \Delta V$ are also called as pre-integrated measurements and they are only affected by $\mathcal{I}_{Wm}, \mathcal{I}_{am}$ and b_g, b_a .

$$\begin{aligned}\Delta R &\triangleq \frac{\mathcal{I}_k}{\mathcal{I}_{k+1}} R &= \frac{\mathcal{I}_k}{\mathcal{I}_k} R^T \frac{\mathcal{I}}{\mathcal{I}_{k+1}} R \\ \Delta P &\triangleq \int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} \frac{\mathcal{I}_k}{\mathcal{I}_s} R^T s_a ds d\tau &= \frac{\mathcal{I}_k}{\mathcal{I}_k} R^T \left(\mathcal{I}_{p_{k+1}} - \mathcal{I}_{p_k} - \mathcal{I}_{v_k} \delta t - \frac{1}{2} \mathcal{I}_g \delta t^2 \right) \\ \Delta V &\triangleq \int_{t_k}^{t_{k+1}} \frac{\mathcal{I}_k}{\mathcal{I}_s} R^T v_a d\tau &= \frac{\mathcal{I}_k}{\mathcal{I}_k} R^T \left(\mathcal{I}_{v_{k+1}} - \mathcal{I}_{v_k} - \mathcal{I}_g \delta t \right) \\ \Delta b_g &\triangleq \int_{t_k}^{t_{k+1}} n_{ug} d\tau &= b_{g_{k+1}} - b_{g_k} \\ \Delta b_a &\triangleq \int_{t_k}^{t_{k+1}} n_{wad} d\tau &= b_{a_{k+1}} - b_{a_k}\end{aligned}$$

so the IMU integration/propagation and pre-integration are connected.

② IMU pre-integration in Manifold (Forster, Luca Carlone), } For example:
VINS-mono (Tong Qin, Shaojie Shen)

Assume $\mathcal{I}_W$ and $\mathcal{I}_a$ is constant in $[t_k, t_{k+1}]$, and $\delta t = t_{k+1} - t_k$ is small.

$$\begin{aligned}\Delta R &\triangleq \frac{\mathcal{I}_k}{\mathcal{I}_{k+1}} R \approx \text{Exp}(\mathcal{I}_W \delta t) & \frac{\mathcal{I}_k}{\mathcal{I}_{k+1}} R \approx I \text{ if } \delta t \rightarrow 0 \\ \Delta P &= \int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} \frac{\mathcal{I}_k}{\mathcal{I}_s} R^T s_a ds d\tau \approx \int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} I_s \cdot \mathcal{I}_a ds d\tau = \frac{1}{2} \mathcal{I}_a \delta t^2 \\ \Delta V &= \int_{t_k}^{t_{k+1}} \frac{\mathcal{I}_k}{\mathcal{I}_s} R^T v_a d\tau \approx \mathcal{I}_a \delta t & \frac{\mathcal{I}_k}{\mathcal{I}_{k+1}} R \approx I \text{ if } \delta t \rightarrow 0\end{aligned}$$

Then: $\left(\frac{\mathcal{I}_{k+1}}{\mathcal{I}_k} R = \frac{\mathcal{I}_k}{\mathcal{I}_k} R \cdot \text{Exp}(\mathcal{I}_W \delta t) \right)$

$\dots$

$$\begin{cases} p_{zk+1} = p_{zk} + q_{vzk} \delta t + \frac{1}{2} \dot{z}_k R^T \ddot{a} \delta t^2 + \frac{1}{6} q_{g0} \delta t^3 \\ q_{vzk+1} = q_{vzk} + \frac{q_{z_k}}{z_k} R^T \ddot{a} \delta t + q_{g0} \delta t \\ b_{gk+1} = b_{gk} + \int_{t_k}^{t_{k+1}} n_{wg} dt \\ b_{ak+1} = b_{ak} + \int_{t_k}^{t_{k+1}} n_{wa} dt \end{cases}$$

This is easy and simple, but might lose some accuracy for fast rotation and low sampling rate (δt is larger)

③ ACI²: analytic combine imu integration : Yulin / chuchu
 CP²: continuous pre-integration : Kevin / Patrick

If we assume $\dot{z}_k w$ and $\dot{z}_k a$ is constant in $[t_k, t_{k+1}]$, then we have:

$$\Delta R \stackrel{\Delta}{=} \frac{z_k}{z_{k+1}} R = \text{Exp}(\dot{z}_k w \cdot \delta t)$$

$$\Delta p \stackrel{\Delta}{=} \int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} \frac{z_k}{z_s} R^T \dot{z}_s a ds d\tau = \int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} \text{Exp}(\dot{z}_k w \delta s) \dot{z}_k a ds d\tau$$

$$\Delta v \stackrel{\Delta}{=} \int_{t_k}^{t_{k+1}} \frac{z_k}{z_\tau} R^T \dot{z}_\tau a d\tau = \int_{t_k}^{t_{k+1}} \text{Exp}(\dot{z}_k w \delta \tau) \dot{z}_k a d\tau$$

The input: $\dot{z}_w = \dot{z}_{wm} - b_{\dot{g}} - n_{\dot{g}} = \underbrace{\dot{z}_{wm} - \hat{b}_{\dot{g}}}_{\dot{z}_{\hat{w}}} - \underbrace{\tilde{b}_{\dot{g}} - n_{\dot{g}}}_{\dot{z}_{\tilde{w}}} \Rightarrow$ we might discretize $n_{\dot{g}}$ as $n_{\dot{g}d}$

$$= \dot{z}_{\hat{w}} + \dot{z}_{\tilde{w}}$$

$\dot{z}_a = \dot{z}_{am} - b_{\dot{a}} - n_{\dot{a}} = \underbrace{\dot{z}_{am} - \hat{b}_{\dot{a}}}_{\dot{z}_{\hat{a}}} - \underbrace{\tilde{b}_{\dot{a}} - n_{\dot{a}}}_{\dot{z}_{\tilde{a}}}$ we might discretize $n_{\dot{a}}$ as $n_{\dot{a}d}$

$$= \dot{z}_{\hat{a}} + \dot{z}_{\tilde{a}}$$

From k to $k+1$, $\delta t = t_{k+1} - t_k$

Then let's integrate ΔR , Δp , Δv , $\Delta b_{\dot{g}}$ and $\Delta b_{\dot{a}}$

$$\Delta \cdot \dot{z}_k \wedge \dot{z}_k$$

$$\Delta R = \text{Exp}(\mathbb{I}_k \omega \delta t)$$

$$\Theta = \sigma_R \omega \delta t$$

$$= \text{Exp}((\mathbb{I}_k \hat{\omega} + \mathbb{I}_k \tilde{\omega}) \delta t) = \text{Exp}(\mathbb{I}_k \hat{\omega} \delta t) \text{Exp}(\text{Tr}(\hat{\Theta}) \mathbb{I}_k \tilde{\omega} \delta t)$$

$$= \text{Exp}(\hat{\Theta}) (I + L \text{Tr}(\hat{\Theta}) \delta t \mathbb{I}_k \tilde{\omega})$$

$$\Delta P = \int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} \text{Exp}((\mathbb{I}_k \hat{\omega} + \mathbb{I}_k \tilde{\omega}) \delta s) ds d\tau (\mathbb{I}_k \hat{a} + \mathbb{I}_k \tilde{a})$$

$$= \int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} \text{Exp}(\mathbb{I}_k \hat{\omega} \delta s) (I + L \text{Tr}(\mathbb{I}_k \hat{\omega} \delta s) \delta s \mathbb{I}_k \tilde{\omega}) ds d\tau (\mathbb{I}_k \hat{a} + \mathbb{I}_k \tilde{a})$$

$$= \underbrace{\int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} \text{Exp}(\mathbb{I}_k \hat{\omega} \delta s) ds d\tau \mathbb{I}_k \hat{a}}_{\Delta \hat{P}} + \underbrace{\int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} \text{Exp}(\mathbb{I}_k \hat{\omega} \delta s) ds d\tau \mathbb{I}_k \tilde{a}}_{\Xi_2} - \underbrace{\int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} \text{Exp}(\mathbb{I}_k \hat{\omega} \delta s) L \mathbb{I}_k \hat{a} L \text{Tr}(\mathbb{I}_k \hat{\omega} \delta s) \delta s ds d\tau \mathbb{I}_k \tilde{\omega}}_{\Xi_4}$$

$$= \Delta \hat{P} + \Xi_2 \mathbb{I}_k \tilde{a} - \Xi_4 \mathbb{I}_k \tilde{\omega}$$

$$\Delta V = \int_{t_k}^{t_{k+1}} \text{Exp}((\mathbb{I}_k \hat{w} + \mathbb{I}_k \tilde{w}) \delta t) d\epsilon d\tau (\mathbb{I}_k \hat{a} + \mathbb{I}_k \tilde{a})$$

$$= \int_{t_k}^{t_{k+1}} \text{Exp}(\mathbb{I}_k \hat{w} \delta t) \cdot (I + L \text{Tr}(\mathbb{I}_k \tilde{w} \delta t) \delta t \mathbb{I}_k \tilde{w} L^T) d\tau (\mathbb{I}_k \hat{a} + \mathbb{I}_k \tilde{a})$$

$$\stackrel{\text{red box}}{=} \underbrace{\int_{t_k}^{t_{k+1}} \text{Exp}(\mathbb{I}_k \hat{w} \delta t) d\tau \mathbb{I}_k \hat{a}}_{\Delta \hat{V}} + \underbrace{\int_{t_k}^{t_{k+1}} \text{Exp}(\mathbb{I}_k \hat{w} \delta t) d\tau}_{\Xi_1} \mathbb{I}_k \tilde{a} - \underbrace{\int_{t_k}^{t_{k+1}} \text{Exp}(\mathbb{I}_k \hat{w} \delta t) L \mathbb{I}_k \hat{a} \text{Tr}(\mathbb{I}_k \tilde{w} \delta t) \delta t \mathbb{I}_k \tilde{w}}_{\Xi_3} d\tau \mathbb{I}_k \tilde{w}$$

$$= \Delta \hat{V} + \Xi_1 \mathbb{I}_k \tilde{a} - \Xi_3 \mathbb{I}_k \tilde{w}$$

Then we have:

$$\begin{cases} \hat{g}_{\mathbb{I}_{k+1}} R = \hat{g}_{\mathbb{I}_k} R \cdot \text{Exp}(\mathbb{I}_k \hat{w} \delta t) \cdot \text{Exp}(\text{Tr}(\hat{\theta}) \delta t \mathbb{I}_k \tilde{w}) \\ \hat{g}_{P_{k+1}} = \hat{g}_{P_k} + \hat{g}_{V_k} \delta t + \hat{g}_{\mathbb{I}_k} R (\Delta \hat{P} + \Xi_2 \mathbb{I}_k \tilde{a} - \Xi_4 \mathbb{I}_k \tilde{w}) + \frac{1}{2} \hat{g}_g \delta t^2 \\ \hat{g}_{V_{k+1}} = \hat{g}_{V_k} + \hat{g}_{\mathbb{I}_k} R (\Delta \hat{V} + \Xi_1 \mathbb{I}_k \tilde{a} - \Xi_3 \mathbb{I}_k \tilde{w}) + \hat{g}_g \delta t \\ \hat{b}_{g_{k+1}} = \hat{b}_{g_k} + \eta_{\text{ugd}} \delta t \\ \hat{b}_{a_{k+1}} = \hat{b}_{a_k} + \eta_{\text{uad}} \delta t \end{cases}$$

$$\text{First: } \begin{cases} \hat{g}_{\mathbb{I}_{k+1}} \hat{R} = \hat{g}_{\mathbb{I}_k} \hat{R} \cdot \text{Exp}(\mathbb{I}_k \hat{w} \delta t) \\ \hat{g}_{P_{k+1}} = \hat{g}_{P_k} + \hat{g}_{V_k} \delta t + \hat{g}_{\mathbb{I}_k} \hat{R} \Delta \hat{P} + \frac{1}{2} \hat{g}_g \delta t^2 \\ \hat{g}_{V_{k+1}} = \hat{g}_{V_k} + \hat{g}_{\mathbb{I}_k} \hat{R} \Delta \hat{V} + \hat{g}_g \delta t \\ \hat{b}_{g_{k+1}} = \hat{b}_{g_k} \\ \hat{b}_{a_{k+1}} = \hat{b}_{a_k} \end{cases}$$

$$\hat{x}_k \rightarrow \hat{x}_{k+1} \quad \text{solved.}$$

solved

Second: compute the linearization for covariance propagation.

Use $\mathbf{SO}(3) + \mathbf{R}^3 + \mathbf{R}^3 + \mathbf{R}^3 + \mathbf{R}^3$ for example:

$$\mathbf{g}_{\mathbf{I}_{k+1}} \mathbf{R} = \mathbf{g}_{\mathbf{I}_{k+1}} \hat{\mathbf{R}} \exp(\delta \mathbf{0}_{k+1}) = \mathbf{g}_{\mathbf{I}_k} \hat{\mathbf{R}} \exp(\delta \mathbf{0}_k) \exp(\mathbf{I}_k \hat{\boldsymbol{\omega}} \delta t) \exp(\mathbf{J}_r(\mathbf{I}_k \hat{\boldsymbol{\omega}} \delta t) \mathbf{I}_k \tilde{\mathbf{w}} \delta t)$$

$$\delta \mathbf{0}_{k+1} = \mathbf{I}_k \hat{\mathbf{R}}^T \delta \mathbf{0}_k + \mathbf{J}_r(\mathbf{I}_k \hat{\boldsymbol{\omega}} \delta t) \delta t \mathbf{I}_k \tilde{\mathbf{w}}$$

$$\mathbf{g}_{\tilde{\mathbf{p}}_{k+1}} = \mathbf{g}_{\tilde{\mathbf{p}}_k} + \mathbf{g}_{\tilde{\mathbf{v}}_k} \delta t - \mathbf{g}_{\mathbf{I}_k} \hat{\mathbf{R}} \mathbf{L} \hat{\mathbf{p}}_{\mathbf{I}_k} \delta \mathbf{0}_k + \mathbf{g}_{\mathbf{I}_k} \hat{\mathbf{R}} (\mathbf{I}_k \tilde{\mathbf{a}} - \mathbf{I}_k \tilde{\mathbf{w}})$$

$$\mathbf{g}_{\tilde{\mathbf{v}}_{k+1}} = \mathbf{g}_{\tilde{\mathbf{v}}_k} - \mathbf{g}_{\mathbf{I}_k} \hat{\mathbf{R}} \mathbf{L} \hat{\mathbf{v}}_{\mathbf{I}_k} \delta \mathbf{0}_k + \mathbf{g}_{\mathbf{I}_k} \hat{\mathbf{R}} (\mathbf{I}_k \tilde{\mathbf{a}} - \mathbf{I}_k \tilde{\mathbf{w}})$$

$$\tilde{\mathbf{b}}_{\mathbf{g}_{k+1}} = \tilde{\mathbf{b}}_{\mathbf{g}_k} + \mathbf{n}_{\mathbf{ngd}} \delta t$$

$$\tilde{\mathbf{b}}_{\mathbf{a}_{k+1}} = \tilde{\mathbf{b}}_{\mathbf{a}_k} + \mathbf{n}_{\mathbf{wad}} \delta t$$

$$\begin{cases} \mathbf{I}_k \tilde{\mathbf{a}} = -\tilde{\mathbf{b}}_{\mathbf{a}} - \tilde{\mathbf{n}}_{\mathbf{ad}} \\ \mathbf{I}_k \tilde{\mathbf{w}} = -\tilde{\mathbf{b}}_{\mathbf{g}} - \mathbf{n}_{\mathbf{gd}} \end{cases}$$

Hence:

$$\underbrace{\begin{bmatrix} \delta \mathbf{0}_{k+1} \\ \mathbf{g}_{\tilde{\mathbf{p}}_{k+1}} \\ \mathbf{g}_{\tilde{\mathbf{v}}_{k+1}} \\ \tilde{\mathbf{b}}_{\mathbf{g}_{k+1}} \\ \tilde{\mathbf{b}}_{\mathbf{a}_{k+1}} \end{bmatrix}}_{\tilde{\mathbf{z}}_{k+1}} = \underbrace{\begin{bmatrix} \mathbf{I}_k \hat{\mathbf{R}}^T & \mathbf{0}_3 & \mathbf{0}_3 & -\mathbf{J}_r(\mathbf{I}_k \hat{\boldsymbol{\omega}} \delta t) \delta t & \mathbf{0}_3 \\ -\mathbf{g}_{\mathbf{I}_k} \hat{\mathbf{R}} \mathbf{L} \hat{\mathbf{p}}_{\mathbf{I}_k} & \mathbf{I}_3 & \mathbf{I}_3 \delta t & \mathbf{g}_{\mathbf{I}_k} \hat{\mathbf{R}} \mathbf{I}_k \tilde{\mathbf{a}} & -\mathbf{g}_{\mathbf{I}_k} \hat{\mathbf{R}} \mathbf{I}_k \tilde{\mathbf{w}} \\ -\mathbf{g}_{\mathbf{I}_k} \hat{\mathbf{R}} \mathbf{L} \hat{\mathbf{v}}_{\mathbf{I}_k} & \mathbf{0}_3 & \mathbf{I}_3 & \mathbf{g}_{\mathbf{I}_k} \hat{\mathbf{R}} \mathbf{I}_k \tilde{\mathbf{a}} & -\mathbf{g}_{\mathbf{I}_k} \hat{\mathbf{R}} \mathbf{I}_k \tilde{\mathbf{w}} \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3 \end{bmatrix}}_{\Phi(\mathbf{I}_{k+1}, \mathbf{I}_k)} \underbrace{\begin{bmatrix} \delta \mathbf{0}_k \\ \mathbf{g}_{\tilde{\mathbf{p}}_k} \\ \mathbf{g}_{\tilde{\mathbf{v}}_k} \\ \tilde{\mathbf{b}}_{\mathbf{g}_k} \\ \tilde{\mathbf{b}}_{\mathbf{a}_k} \end{bmatrix}}_{\tilde{\mathbf{z}}_k} + \underbrace{\begin{bmatrix} -\mathbf{J}_r(\mathbf{I}_k \hat{\boldsymbol{\omega}} \delta t) \delta t & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 \\ \mathbf{g}_{\mathbf{I}_k} \hat{\mathbf{R}} \mathbf{I}_k \tilde{\mathbf{a}} & -\mathbf{g}_{\mathbf{I}_k} \hat{\mathbf{R}} \mathbf{I}_k \tilde{\mathbf{w}} & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 \\ \mathbf{g}_{\mathbf{I}_k} \hat{\mathbf{R}} \mathbf{I}_k \tilde{\mathbf{a}} & -\mathbf{g}_{\mathbf{I}_k} \hat{\mathbf{R}} \mathbf{I}_k \tilde{\mathbf{w}} & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3 \delta t & \mathbf{0}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3 \delta t & \mathbf{0}_3 \end{bmatrix}}_{\mathbf{0}} \underbrace{\begin{bmatrix} \mathbf{n}_{\mathbf{gd}} \\ \mathbf{n}_{\mathbf{ad}} \\ \mathbf{n}_{\mathbf{ugd}} \\ \mathbf{n}_{\mathbf{wad}} \end{bmatrix}}_{\mathbf{0}}$$

$$\tilde{x}_{k+1} = \tilde{\Phi}(k+1, k) \tilde{x}_k + G_k n_{2d}$$

$$\text{Hence: } P_{k+1} = \tilde{\Phi}(k+1, k) P_k \tilde{\Phi}^T(k+1, k) + G_k Q_{2d} G_k^T$$

solved

now The linearisation using $SE_2(3)$

The states: $SE_2(3) + R^3 + R^3$

$$x_1 = (x_n, x_b) \quad x_n = (\frac{g}{I_k} R, g p_2, g v_2) = \begin{bmatrix} \frac{g}{I_k} R & g p_2 & g v_2 \\ 0_{1 \times 3} & 1 & 0 \\ 0_{1 \times 3} & 0 & 1 \end{bmatrix}, \quad x_b = (bg, ba)$$

Let's define the error states

$$x_1 = \hat{x}_1 \oplus \delta x_1 = (\exp(\delta x_n) \hat{x}_n, \hat{x}_b + \delta x_b)$$

$$= \begin{bmatrix} \exp(\delta \theta) \frac{g}{I_k} \hat{R} \\ \exp(\delta \theta) g \hat{p}_2 + J_{v_2}(\delta \theta_2) \delta p \\ \exp(\delta \theta) g \hat{v}_2 + J_{v_2}(\delta \theta_2) \delta v \\ \hat{bg} + \delta bg \\ \hat{ba} + \delta ba \end{bmatrix}$$

Then:

$$\left\{ \begin{array}{l} \frac{g}{I_{k+1}} R = \frac{g}{I_k} R - \frac{I_k}{I_{k+1}} R \\ g p_{2k+1} = g p_{2k} + g v_{2k} \delta t + \frac{g}{I_k} R \Delta p + \frac{1}{2} g g \delta t^2 \\ g v_{2k+1} = g v_{2k} + \frac{g}{I_k} R \Delta v + g g \delta t \\ bg_{k+1} = bg_k + \omega_{bg} \delta t \\ ba_{k+1} = ba_k + \omega_{ba} \delta t \end{array} \right.$$

$$x_{n_{k+1}} = \Gamma_k \cdot \Psi(x_{n_k}) \cdot \gamma_k$$

$$= \underbrace{\begin{bmatrix} I_3 & \frac{1}{2} g g \delta t^2 & g \delta t \\ 0_{1 \times 3} & 1 & 0 \\ 0_{1 \times 3} & 0 & 1 \end{bmatrix}}_{\Gamma_k} \underbrace{\begin{bmatrix} \frac{g}{I_k} R & g p_{2k} + g v_{2k} \delta t & g v_{2k} \\ 0_{1 \times 3} & 1 & 0 \\ 0_{1 \times 3} & 0 & 1 \end{bmatrix}}_{\Psi(x_{n_k})} \underbrace{\begin{bmatrix} \Delta R & \Delta p & \Delta v \\ 0_{1 \times 3} & 1 & 0 \\ 0_{1 \times 3} & 0 & 1 \end{bmatrix}}_{\gamma_k}$$

The properties of $\psi(\cdot)$

$$\begin{cases} \psi(x_{ni} \cdot x_{nj}) = \psi(x_{ni}) \cdot \psi(x_{nj}) \\ \psi(\bar{\text{Exp}}(\delta x_n)) \approx \bar{\text{Exp}}(\bar{F} \delta x_n) \\ \psi(\bar{\text{Exp}}(\delta x_n) \hat{x}_n) = \psi(\bar{\text{Exp}}(\delta x_n)) \cdot \psi(\hat{x}_n) = \bar{\text{Exp}}(\bar{F} \delta x_n) \psi(\hat{x}_n) \end{cases} \quad \bar{F} \triangleq \begin{bmatrix} I_3 & 0_3 & 0_3 \\ 0_3 & I_3 & I_3 \delta t \\ 0_3 & 0_3 & I_3 \end{bmatrix}$$

Hence, we can do the linearization as:

$$\begin{aligned} x_{n_{k+1}} &= \bar{\text{Exp}}(\delta x_{n_{k+1}}) \cdot \hat{x}_{n_{k+1}} \\ &= \bar{I}_k \psi(x_{n_k}) \tilde{Y}_k \\ &= \bar{I}_k \psi(\bar{\text{Exp}}(\delta x_{n_k}) \hat{x}_{n_k}) \hat{Y}_k \cdot \tilde{Y}_k \\ &\approx \bar{I}_k \bar{\text{Exp}}(\bar{F} \delta x_{n_k}) \psi(\hat{x}_{n_k}) \hat{Y}_k \tilde{Y}_k \end{aligned}$$